Practical 01: Hello World Android Application

Source: Android Programming Practical(1).pdf | Source pages: 3–13

Aim

To develop and test an Android application demonstrating hello world android application.

Steps to Perform the Practical

1. Open Android Studio and select New Project.

2. Choose Empty Views Activity, enter project name HelloWorld, select Java, and finish creating the project.

3. Review the Android Studio project panel and locate MainActivity.java, activity_main.xml, AndroidManifest.xml and resource files.

4. Use the supplied MainActivity.java and activity_main.xml code. Check the package name matches your project.

5. Open Device Manager and create/start the AVD described in the practical, or connect an Android device with USB debugging enabled.

6. Click Run. Confirm that Hello, World! appears on the emulator/device.

Result

The application is run and the expected behavior is verified according to the practical instructions and code.

Note: Use the package name and resource names from the source code consistently. If Android Studio reports a missing resource or class, check spelling and ensure all files are in the correct folders.

Practical 01

Develop a simple "Hello World" Android application demonstrating project creation, Android Studio interface, Android components, AVD creation, and application execution.

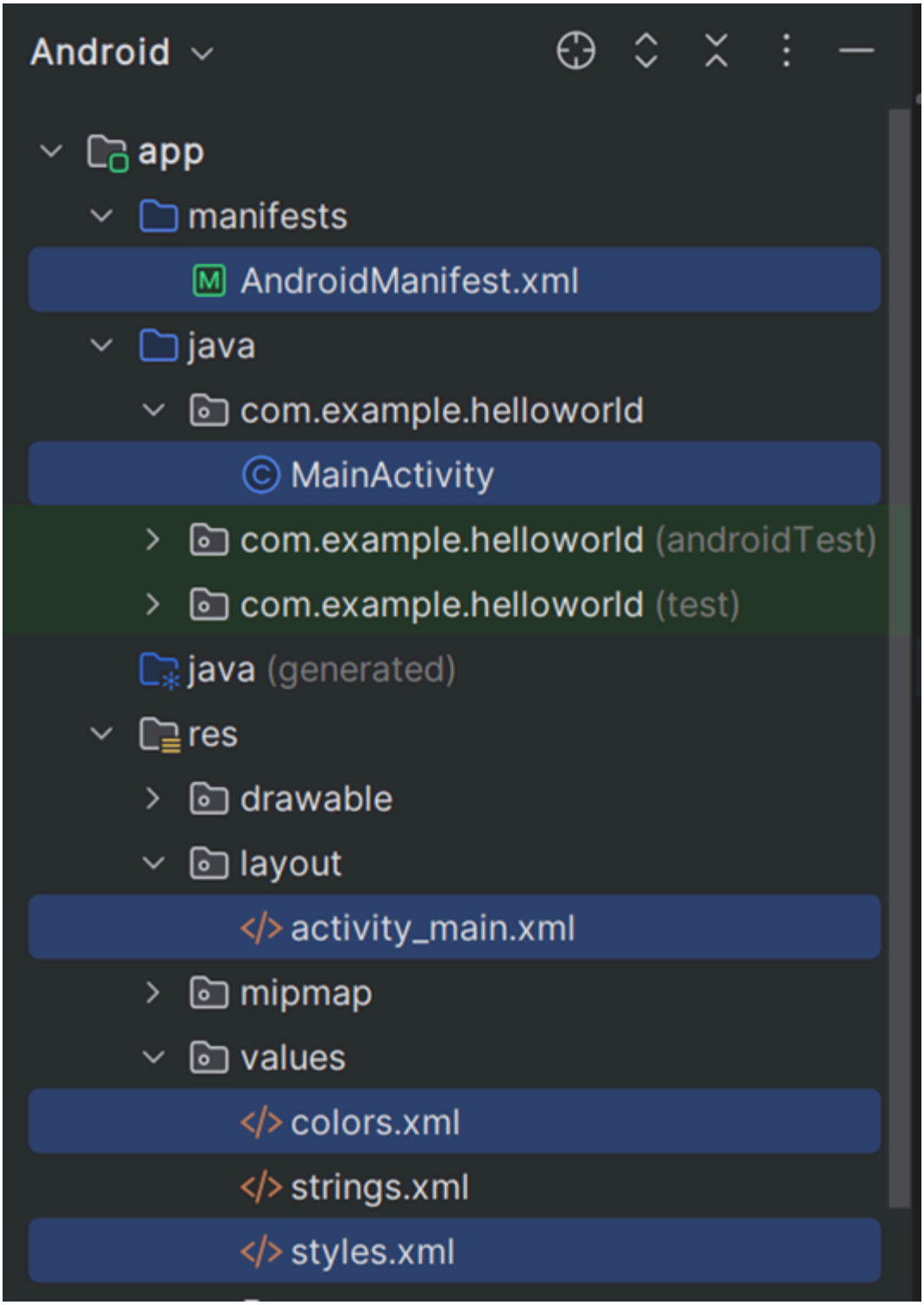
Project 1: Hello World

Project Name: HelloWorld

Template: Empty Views Activity

Project Architecture:

Total 5 files


Android
app
manifests
AndroidManifest.xml
java
com.example.helloworld
MainActivity
com.example.helloworld (androidTest)
com.example.helloworld (test)
java (generated)
res
drawable
layout
activity_main.xml
mipmap
values
colors.xml
strings.xml
styles.xml

I

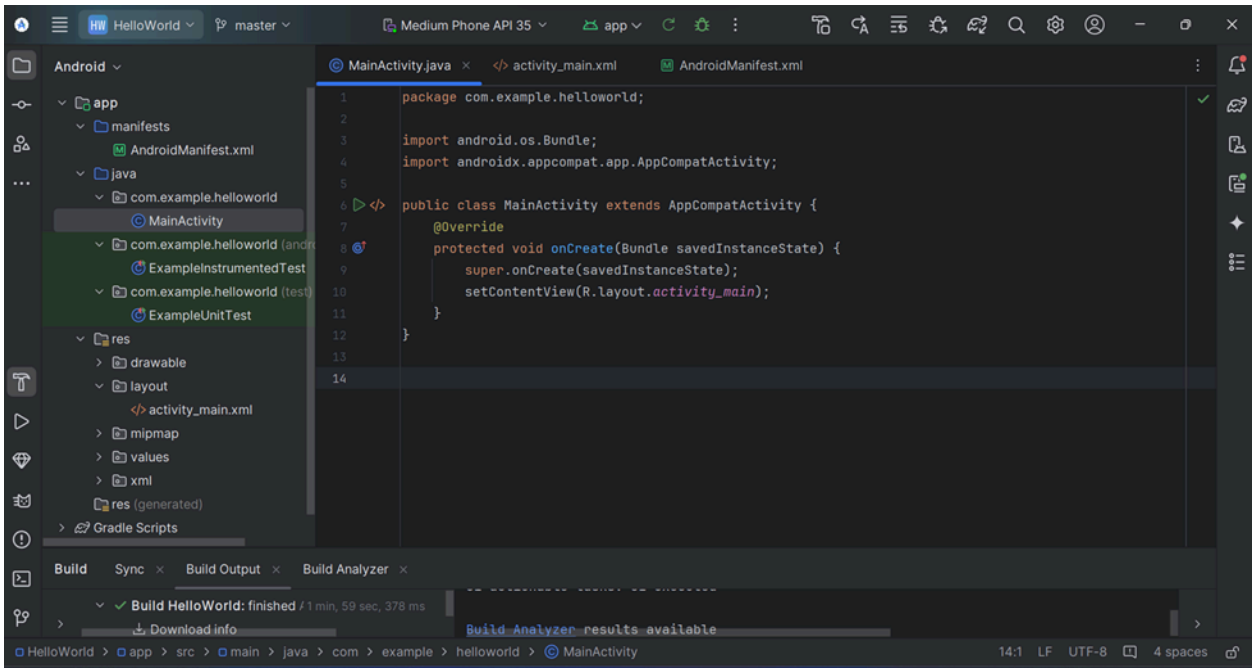
Filename: MainActivity.java

```
package com.example.helloworld;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
      @Override
      protected void onCreate(Bundle savedInstanceState) {
         super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      }
}
```

Path: app>java>com.example.helloworld>MainActivity.java



II

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

        xmlns:android="http://schemas.android.com/apk/res/android"

        android:layout_width="match_parent"

        android:layout_height="match_parent"

        android:orientation="vertical"

        android:gravity="center"

        android:padding="16dp">


        <TextView

        android:id="@+id/text_hello"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Hello, World!"

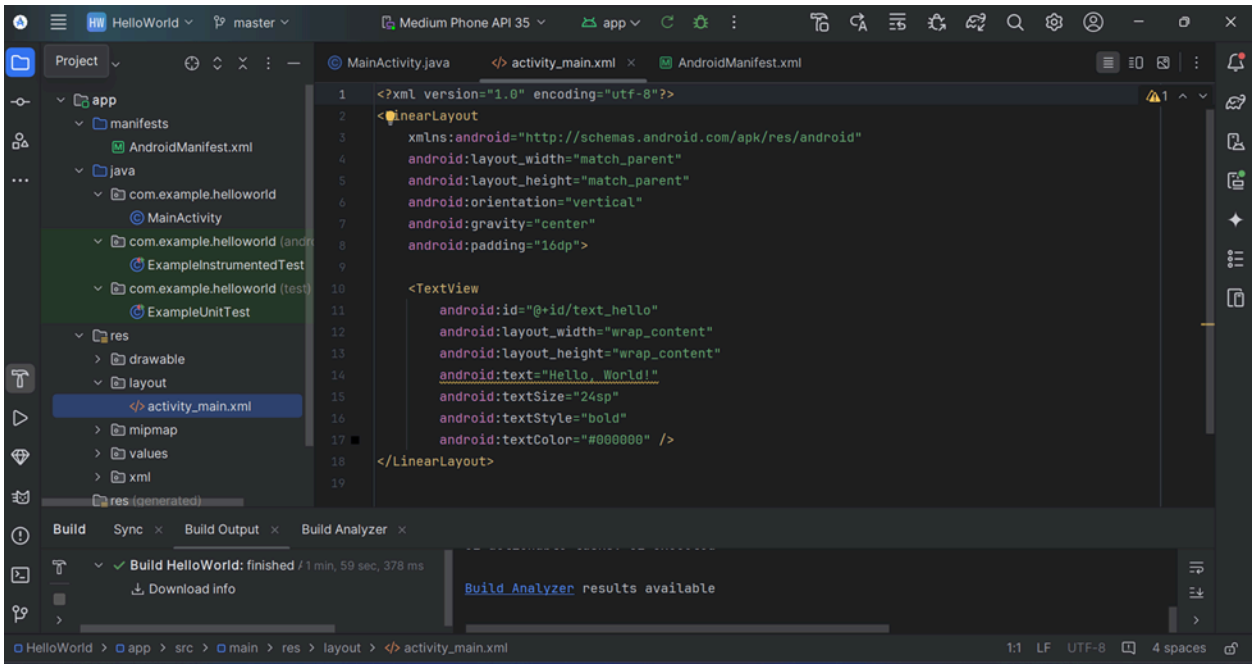
        android:textSize="24sp"

        android:textStyle="bold"

        android:textColor="#000000" />

</LinearLayout>
```

Path: app>res>layout>activity_main.xml



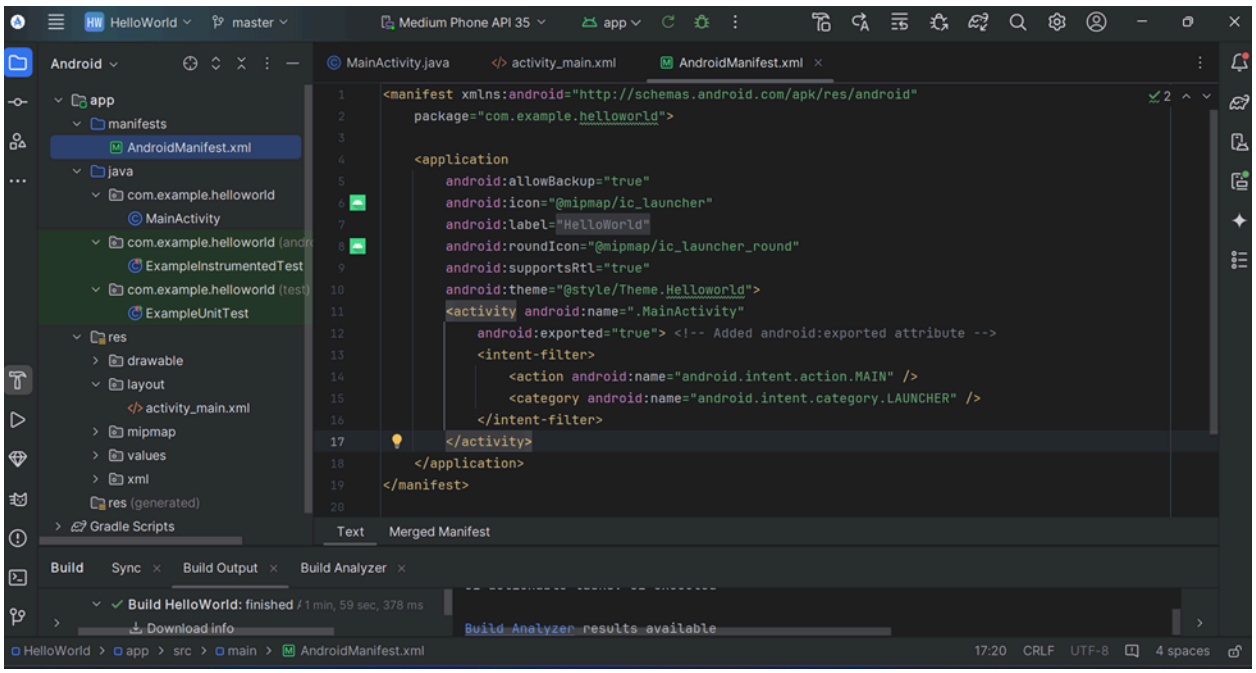
III

Filename: AndroidManifest.xml

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
        package="com.example.helloworld">
        <application
    android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Helloworld">
        <activity android:name=".MainActivity"
        android:exported="true"> <!-- Added android:exported attribute -->
        <intent-filter>
                <action android:name="android.intent.action.MAIN" />
          <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
```

```
        </activity>
        </application>
</manifest>
```

Path: app>manifest>AndroidManifest.xml

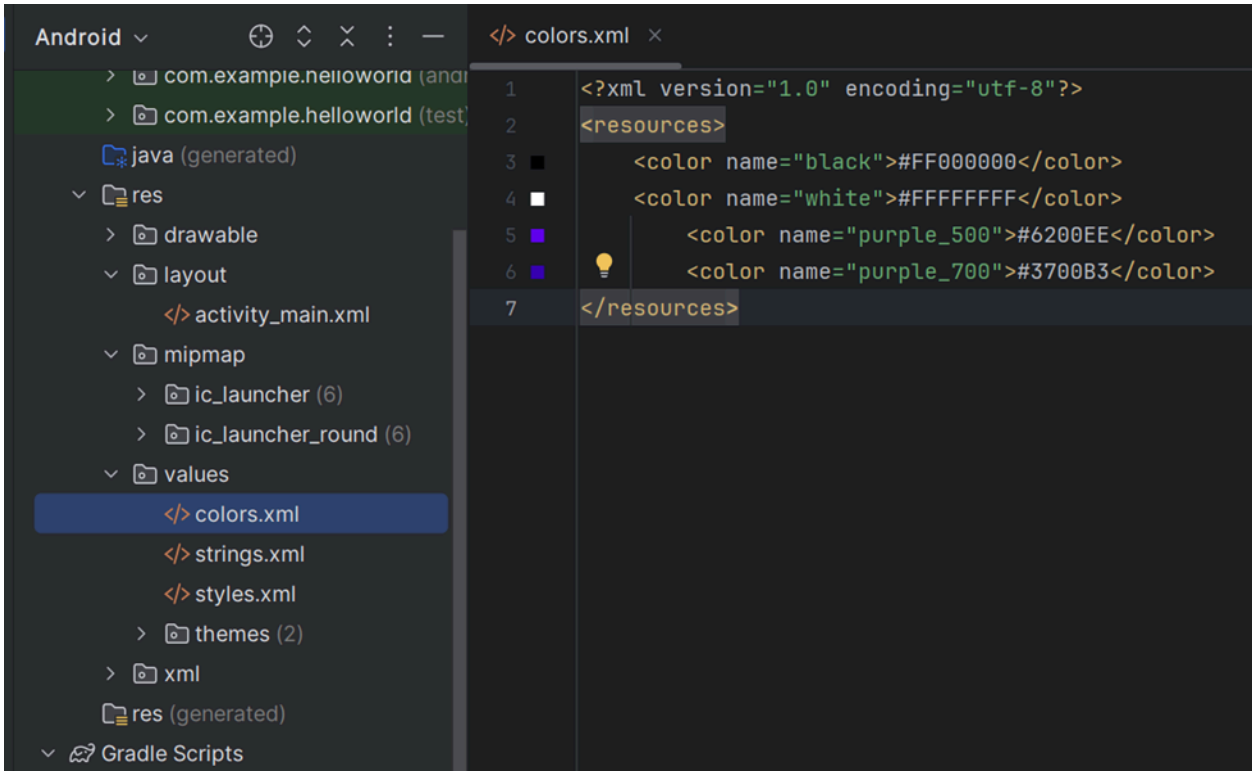


IV

Filename: colors.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <resources>
      <color name="black">#FF000000</color>
      <color name="white">#FFFFFFFF</color>
      <color name="purple_500">#6200EE</color>
      <color name="purple_700">#3700B3</color>
 </resources>
```

Path: app>res>values>colors.xml

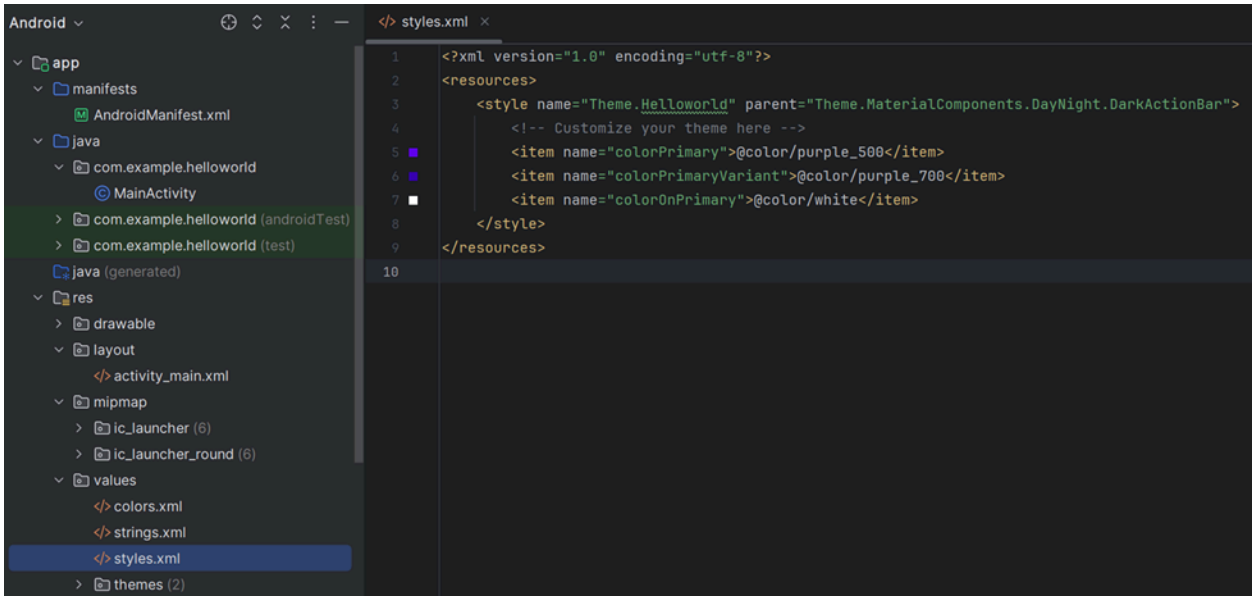


V

Filename: styles.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <resources>
      <style name="Theme.Helloworld"
parent="Theme.MaterialComponents.DayNight.DarkActionBar">
      <!-- Customize your theme here -->
      <item name="colorPrimary">@color/purple_500</item>
      <item name="colorPrimaryVariant">@color/purple_700</item>
      <item name="colorOnPrimary">@color/white</item>
      </style>
 </resources>
```

Path: app>res>values>styles.xml



VI (Prebuild Project Code: To ensure proper project detail: No need to update until needed)

Filename: build.gradle.kts (Module:app)

```
plugins {
      alias(libs.plugins.android.application)
 }

 android {
      namespace = "com.example.helloworld"
      compileSdk = 35

      defaultConfig {
      applicationId = "com.example.helloworld"
      minSdk = 24
      targetSdk = 35
      versionCode = 1
      versionName = "1.0"

      testInstrumentationRunner = "androidx.test.runner.AndroidJUnitRunner"
      }

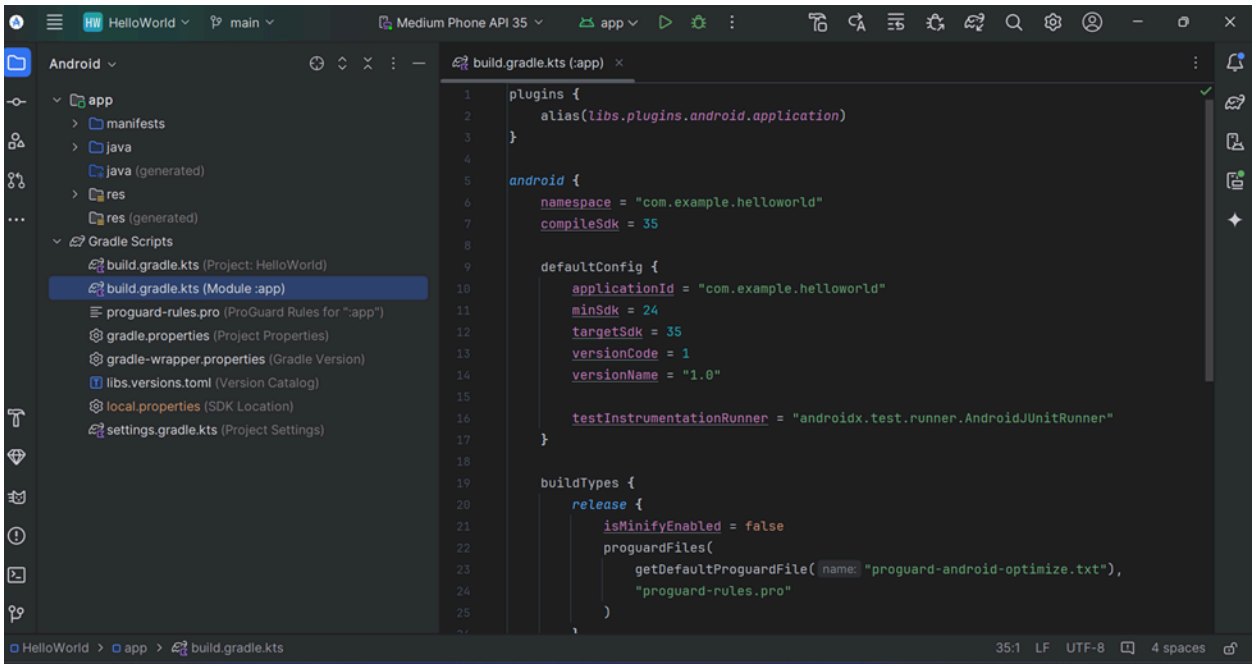
      buildTypes {
      release {
            isMinifyEnabled = false
            proguardFiles(
                  getDefaultProguardFile("proguard-android-optimize.txt"),
                  "proguard-rules.pro"
            )
      }
```

```
        }
        compileOptions {
        sourceCompatibility = JavaVersion.VERSION_11
        targetCompatibility = JavaVersion.VERSION_11
        }
}

dependencies {

        implementation(libs.appcompat)
        implementation(libs.material)
        implementation(libs.activity)
        implementation(libs.constraintlayout)
        testImplementation(libs.junit)
        androidTestImplementation(libs.ext.junit)
        androidTestImplementation(libs.espresso.core)
}
```

Path: Gradle Scripts > build.gradle.kts

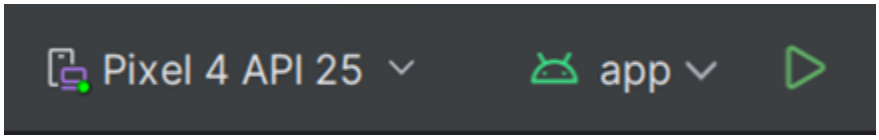


```
        }
    }
    compileOptions {
        sourceCompatibility = JavaVersion.VERSION_11
        targetCompatibility = JavaVersion.VERSION_11
    }
}

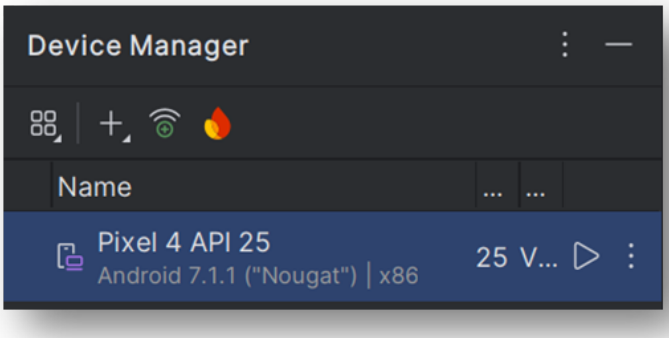
dependencies {

    implementation(libs.appcompat)
    implementation(libs.material)
    implementation(libs.activity)
    implementation(libs.constraintlayout)
    testImplementation(libs.junit)
    androidTestImplementation(libs.ext.junit)
    androidTestImplementation(libs.espresso.core)
}
```

Above the Android Studio near Main Menu Click the Play button to Run the Program



Recommended compulsory AVD Emulator: Pixel 4 API 25 with version 7.1.1 Nougat



Output:

